

Overview

Part 01 —  Machine Learning Foundations · 01-ml-foundations/README.md

Module 01 — Machine Learning Foundations

Where you are: Stop 1 of 16 on the end-to-end AI track. **Time:** ~20–25 hours · **Prereq:** Python, high-school algebra, willingness to meet calculus again.

Everything later in this track — transformers, agents, RAG, evals — is machine learning wearing a bigger coat. If you can explain *bias/variance*, *gradient descent*, and *why your validation split lied to you*, the rest of the track reads like engineering. If you can't, it reads like magic.

Learning objectives

By the end of this module you can:

1. Frame a business problem as a supervised, unsupervised, or reinforcement learning task — and say when **not** to use ML.
2. Derive and implement linear & logistic regression with gradient descent from scratch (NumPy only).
3. Explain the bias–variance decomposition and diagnose under/overfitting from learning curves.
4. Build an honest evaluation setup: train/val/test, cross-validation, leakage hunting, and the right metric for the cost of being wrong.
5. Use tree ensembles (Random Forest, gradient boosting) and explain why they still beat deep learning on tabular data.
6. Read a model card / experiment report critically.

Suggested path

#	Do this	File	Time
1	Read the core concepts	notes/01-core-concepts.md	3h
2	Work the math of learning	notes/02-optimization-and-generalization.md	3h
3	Run the from-scratch trainer	code/linear_models_from_scratch.py	2h
4	Run the honest-evaluation lab	code/evaluation_playbook.py	2h
5	Present it back	slides/ (ml-foundations.pptx)	1h
6	Do the exercises	lab/EXERCISES.md	6h
7	Read 2 papers	papers/PAPERS.md	4h

The 12 terms you must own

feature · label · loss function · gradient descent · learning rate · overfitting · regularization · bias-variance · cross-validation · data leakage · class imbalance · ensemble

Exit check

You can hand a colleague a notebook that loads a tabular dataset, produces a baseline, a tuned model, a leakage-free CV score, a calibration curve, and one paragraph on when the model should *not* be trusted.